

# Ridwan Olalekan Olabiyi

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## Research Interests

Scientific machine learning; Bayesian inference; stochastic inverse problems; uncertainty quantification; governing-physics discovery; causal discovery; materials characterization; neural/operator learning; image-based 3D reconstruction; photogrammetry; photometric stereo; statistical quality engineering; advanced manufacturing.

## Education

**Ph.D., Industrial Engineering, Arizona State University** Expected Spring 2027

Advisor: Ashif Iqbal.

Working dissertation title: *Physics-Based Machine Learning for Inverse Problems and Scientific Discovery under Uncertainty*.

**M.S., Industrial Engineering, Arizona State University** December 2024

Advisor: Ashif Iqbal. Degree GPA: 3.73.

Thesis: *A Bayesian Spatiotemporal Modeling Approach to the Inverse Heat Conduction Problem*.

**B.Sc., Agricultural & Environmental Engineering, Obafemi Awolowo University** 2019

First-Class Honors; CGPA 4.58/5.00; class rank 2/25.

**Venture in Management Programme, Lagos Business School, Pan-Atlantic University** 2020

## Academic Appointments & Research Experience

**Research Associate, School of Computing and Augmented Intelligence, Arizona State University** Current

Develop physics-aware and uncertainty-aware computational methods for inverse problems, scientific discovery, and computationally efficient engineering modeling. Current work spans advanced manufacturing, materials characterization, dynamical systems, scientific machine learning, image-based quality assurance, and collaborations with researchers at NIST and Oak Ridge National Laboratory.

**Graduate Research Associate, Arizona State University** 2021–2022

Research under Ashif Iqbal in Bayesian inference, inverse problems, data-driven engineering, and scientific machine learning.

**Graduate Teaching Assistant, Arizona State University** 2022–2023

Supported IEE 380, *Probability and Statistics for Engineering Problem Solving*, in Fall 2022 and Spring 2023.

## Selected Research Contributions

- **Physics discovery under uncertainty:** Developed a stochastic inverse framework for recovering governing equations in the presence of measurement noise and intrinsic parameter variability; published in *Communications Physics* (2026).
- **Bayesian spatiotemporal inverse heat conduction:** Developed an uncertainty-aware framework for reconstructing latent thermal quantities from indirect temperature measurements; conference publication (2023) and journal publication (2024).
- **Stochastic inverse indentation:** Developed inverse methodologies for elastoplastic property recovery from indentation data; current extension addresses manufacturing-induced variability and measurement uncertainty with NIST/ORNL collaborators.
- **CRONet and topology optimization:** Developed neural/operator-learning approaches for accelerating iterative topology optimization; published in *Manufacturing Letters* and extended through FPGA/AIE-ML acceleration research at IEEE FCCM 2026.
- **Image-based 3D reconstruction for AM quality assurance:** Developing a non-contact inspection system for additively manufactured components that combines controlled image acquisition, multi-view photogrammetry, photometric stereo, calibration, surface reconstruction, point-cloud fusion, and CAD/reference comparison. Current work focuses on robust reconstruction and positive surface-deviation detection; longer-term integration targets post-print inspection, in-situ quality monitoring, and targeted surface processing on a Meltio robotic metal-AM platform.
- **Causal discovery:** Ongoing work on nonlinear dynamical-system discovery through counterfactual explanations and causal reasoning.

## Journal Publications

- Christy Dunlap, **Ridwan Olabiyi**, Firas Al-Hindawi, Hari Pandey, Stephen Pierson, Daniel Curl, Braden Stevens, Mohammad Ishraq Hossain, Annapurna Parjuli, Chinmaya Joshi, Ashif Iquebal, and Han Hu. “Open datasets and machine learning for two-phase heat transfer: a review following a spatial-temporal taxonomy.” *Transport Phenomena*, 2026. doi:10.1515/tp-2026-0081.
- **Ridwan Olabiyi**, Han Hu, and Ashif Iquebal. “Discovering Governing Physics in the Presence of Uncertainty.” *Communications Physics*, 2026. doi:10.1038/s42005-026-02711-7.
- **Ridwan Olabiyi**, Hui Yang, and Ashif Iquebal. “CRONet: A Convolutional Recurrent Operator Approximator Network to Accelerate Topology Optimization.” *Manufacturing Letters*, 2025. doi:10.1016/j.mfglet.2025.06.125.
- **Ridwan Olabiyi**, Hari Pandey, Han Hu, and Ashif Iquebal. “A Bayesian Spatiotemporal Modeling Approach to the Inverse Heat Conduction Problem.” *ASME Journal of Heat and Mass Transfer*, 146(9), 091403, 2024. doi:10.1115/1.4065451.

## Peer-Reviewed Conference Publications

- Kaustubh Mhatre, Vedant Tewari, Aditya Ray, Farhan Khan, **Ridwan Olabiyi**, Ashif Iquebal, and Aman Arora. “Accelerating Topology Optimization on AMD Versal AIE-ML Engines.” *2026 IEEE 34th Annual International Symposium on Field-Programmable Custom Computing Machines (FCCM)*, 2026. doi:10.1109/FCCM68464.2026.00047.
- **Ridwan Olabiyi**, Jordan Weaver, and Ashif Iquebal. “Characterization of Elastoplastic Properties of Additively Manufactured Specimens from Indentation Data Using Stochastic Inverse Modeling.” *ASME 2024 International Manufacturing Science and Engineering Conference*, 2024. doi:10.1115/MSEC2024-125452.
- **Ridwan Olabiyi**, Hari Pandey, Han Hu, and Ashif Iquebal. “A Bayesian Spatio-Temporal Modeling Approach to the Inverse Heat Conduction Problem.” *ASME 2023 Heat Transfer Summer Conference*, 2023. doi:10.1115/HT2023-107671.

## Preprints & Open Review

- **Ridwan Olabiyi**, Ishraq R. Rahman, and Ashif Iquebal. “CaNDiCE: Causal Discovery of Nonlinear Dynamics Through Counterfactual Explanations.” OpenReview, ICLR 2026 submission. OpenReview.
- Kaustubh Mhatre, Vedant Tewari, Aditya Ray, Farhan Khan, **Ridwan Olabiyi**, Ashif Iquebal, and Aman Arora. “Accelerating CRONet on AMD Versal AIE-ML Engines.” arXiv:2604.14700, 2026. arXiv.
- **Ridwan Olabiyi**, Han Hu, and Ashif Iquebal. “Discovering Governing Equations in the Presence of Uncertainty.” arXiv:2507.09740, 2025. arXiv.

## Manuscript in Preparation

- **Ridwan Olabiyi**, Jordan S. Weaver, Patxi Fernandez-Zelaia, and Ashif Iquebal. “Accelerating Elastoplastic Property Characterization Under Manufacturing-Induced Variability Through Stochastic Inverse Modeling of Indentation Data.” Manuscript in preparation for an IISE journal submission, 2026.

## Selected Conference Presentations

- INFORMS Annual Meeting Job Market Showcase, San Francisco, 2026: *Stochastic Inverse Modeling for Rapid Elastoplastic Property Characterization of Additively Manufactured Components*.
- NAMRC 53, Greenville, SC, 2025: *CRONet: A Convolutional Recurrent Operator Approximator Network to Accelerate Topology Optimization*.
- IISE Annual Conference, Atlanta, 2025: stochastic inverse modeling for elastoplastic characterization; technical and poster presentations.
- INFORMS Annual Meeting, Seattle, 2024: *Consistent Discovery of Dynamical Systems Using Stochastic Inverse Modeling*; QSR Best Student Paper finalist presentation.
- ASME MSEC, Knoxville, 2024: stochastic inverse indentation for additively manufactured specimens.
- INFORMS Annual Meeting, Phoenix, 2023: Bayesian spatiotemporal inverse heat-conduction research; technical and poster presentations.
- ASME Heat Transfer Summer Conference, Washington, DC, 2023: Bayesian spatiotemporal inverse heat conduction.
- INFORMS Annual Meeting, Indianapolis, 2022: *Minimal Subset Learning in Federated Smart Manufacturing*.

## Teaching Experience

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| <b>IEE 570 – Advanced Quality Control, Arizona State University</b>                             | Summer 2026 |
| Graduate-level instruction, student consultations, assessment, exam review, and course grading. |             |
| <b>IEE 470 – Quality Control, Arizona State University</b>                                      | Spring 2026 |

Undergraduate quality-control instruction connecting statistical methods with engineering process-improvement decisions.

**IEE 570 – Advanced Quality Control, Arizona State University** Summer 2024

Instructional support in advanced statistical quality-control concepts and applications.

**IEE 380 – Probability and Statistics for Engineering Problem Solving, ASU** Fall 2022, Spring 2023

Graduate Teaching Assistant.

**Volunteer GRE Instructor, Nigeria** Oct. 2020–July 2021

Quantitative reasoning and standardized-test preparation for prospective graduate students.

**Classroom Teacher, City College Enugu, Nigeria** 2020–2021

Mathematics, Physics, Basic Science, and Technology.

**Course Tutor, Brainfill Academy, Nigeria** 2020

Advanced-level mathematics including algebra, trigonometry, geometry, and calculus.

## **Research Mentoring**

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- Mentor a recent Mechanical Engineering graduate and an Industrial Engineering Ph.D. student on an image-based 3D reconstruction system for additive-manufacturing quality assurance. The project combines controlled imaging, photogrammetry, photometric stereo, calibration, surface reconstruction, and reference-geometry comparison, with longer-term integration targeting robotic metal-AM inspection and targeted post-processing; the collaboration has also included award-winning engineering competitions.
- Mentored an international undergraduate researcher from France through ASU's Summer Research Initiative on active-learning research.
- Co-mentored two master's students alongside Dr. Ashif Iquebal on generative CAD generation from hand-drawn sketches and inverse modeling for microstructural characterization using eddy-current field data.
- Provide recurring research discussion, technical feedback, troubleshooting, and methodological guidance to additional master's and Ph.D. researchers.

## **Honors, Awards & Fellowships**

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- First Place, NSF Future Manufacturing Data Challenge, 2026.
- Best Presentation Design Award, NSF Future Manufacturing Data Challenge, 2026.
- Winner, Semiconductor Solutions Challenge, Problem D, 2026.
- Trailblazers in Engineering Fellow, Purdue University College of Engineering, 2025.
- National Science Foundation Travel Award, NAMRC 53 / MSEC 2025.
- Harold & Lucille Dunn Memorial Engineering Scholarship, 2025, 2024, 2023, 2022.
- Graduate College Travel Award, IISE Annual Conference, 2025.
- SCAI IE Doctoral Fellowship, 2025 and 2024.
- Finalist, INFORMS Quality, Statistics & Reliability Best Student Paper Competition, 2024.
- Graduate Student Government Travel Grant, INFORMS Annual Meeting, 2024.
- SCAI Conference Funding, INFORMS Annual Meeting, 2024 and 2023.
- NSF Early Career Researcher Travel Award, MSEC 2024.
- University Graduate Fellowship, Arizona State University, 2022.
- Amazon SCOT / INFORMS Scholarship, 2022.
- CIDSE Doctoral Fellowship, Arizona State University, 2021.
- Male Best Speaker, NYSC Enugu Orientation Camp, 2020.
- Venture in Management Programme Scholarship, Lagos Business School, 2020.
- Overall Best Graduating Student (Male), Obafemi Awolowo University, 2019.
- Best Graduating Student, Agricultural & Environmental Engineering, OAU, 2019.
- Professor V. A. Oyenuga Prize, OAU, 2019.
- Engineer E. A. Ladipo's Prize, OAU, 2019.
- Abdulkabir Aliu Foundation Undergraduate Scholarship, 2016–2018.

## **Conference Leadership & Professional Service**

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- Co-Organizer, invited Data Mining cluster session, *Data-Driven Modeling and Scientific Discovery in Complex Systems*, INFORMS Annual Meeting, 2026.

- Session Chair, INFORMS Annual Meeting, 2025.
- Session Chair, Additive Manufacturing technical session, NAMRC 53 / MSEC, 2025.
- Session Chair, *Data-driven Models for Scientific Discovery*, INFORMS Annual Meeting, 2024.

**Journal Peer Review:** *Computers & Industrial Engineering*; *Engineering Applications of Artificial Intelligence*; *Journal of Manufacturing Processes*; *The Journal of Supercomputing*.

**Conference Peer Review:** ASME MSEC (2024–2025); NAMRC (2024–2025); IISE Annual Conference & Expo (2025).

## Leadership, Outreach & Volunteering

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- Vice President, ASU INFORMS Student Chapter, 2023–2025.
- Resource Person, EducationUSA Graduate Boot Camp, 2021 and 2024.
- Student Volunteer, INFORMS Annual Meeting, Phoenix, 2023.
- Google Road Show Internet-Safety Outreach, Lagos State, 2020.
- Volunteer, Junior Achievement Nigeria, 2019–2020.
- Agro-Allied Community Development Service, 2019–2020.
- Chairman, Best Brain Challenge Committee, Obafemi Awolowo University, 2019.
- Prospective Corps Members Welcoming & Guidance Team, Muslim Corpsers Association of Nigeria, 2019.
- Co-convener, Faculty of Technology Muslim Students Association Annual Symposium, OAU, 2018.
- Departmental Football Coach, OAU, 2017–2018.
- Executive Member, Student Representative Council, AGEESS, OAU, 2015–2018.
- Course Tutor, Muslim Student Society of Nigeria, OAU Chapter, 2014–2016.

## Professional Memberships

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INFORMS; Institute of Industrial and Systems Engineers (IISE); American Society of Mechanical Engineers (ASME); Society of Manufacturing Engineers (SME); Society for Industrial and Applied Mathematics (SIAM); Nigerian Society of Engineers (NSE); Nigerian Institution of Agricultural Engineers (NIAE).

## Technical Skills & Certifications

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**Programming/Software:** Python, MATLAB, Jupyter, Git/GitHub, AMPL, IBM CPLEX, ABAQUS, COMSOL Multiphysics.

**Methods:** Bayesian inference, stochastic inverse modeling, uncertainty quantification, design of experiments, causal inference, scientific machine learning, neural/operator learning, statistical quality control, finite-element modeling, image-based 3D reconstruction, photogrammetry, photometric stereo, point-cloud processing, and CAD-referenced surface inspection.

**CITI Program:** Human Research: IRB – Biomedical Research (Group 1), completed 2026, valid through August 2030; Responsible Conduct of Research, Graduate Student and Postdoctoral Researcher RCR, completed 2024.

## Prior Engineering Experience

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**Assistant Program Engineer (NYSC), Enugu State Agricultural Development Program** 2019–2020  
Supported agricultural-equipment management, technical training for farmers, and engineering-program operations.

**Workshop / Production Engineering Intern, International Institute of Tropical Agriculture** 2017–2018  
Supported maintenance and operation of agricultural machinery, workshop equipment, and production/packaging systems.

## Selected Media & Professional Profiles

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- Purdue University College of Engineering – 2025 Trailblazers in Engineering.
- The Guardian – NSF-supported manufacturing research.
- The Guardian – INFORMS/QSR recognition.
- Amazon Science – 2022 Amazon SCOT / INFORMS Scholars.